

more natural measure of evidence as it treats the null and alternative hypotheses symmetrically and represents the factor by which rational agents should update their beliefs in light of the data. Replication success is ideally evaluated along multiple dimensions, as nicely exemplified by the RPCB, EPRP, and RPP. Replications that are successful on multiple criteria provide more convincing support for the original finding, while replications that are successful on fewer criteria require closer examination. Fortunately, the use of multiple methods is already standard practice in replication assessment, so our proposal to use both of them does not require a major paradigm shift.

While the equivalence test and the Bayes factor are two principled methods for analyzing original and replication studies with null results, they are not the only possible methods for doing so. A straightforward extension would be to first synthesize the original and replication effect estimates with a meta-analysis, and then apply the equivalence and Bayes factor tests to the meta-analytic estimate similar to the meta-analytic non-significance criterion used by the RPCB. This could potentially improve the power of the tests, but consideration must be given to the threshold used for the p -values/Bayes factors, as naive use of the same thresholds as in the standard approaches may make the tests too liberal (*Shun et al., 2005*). Furthermore, there are various advanced methods for quantifying evidence for absent effects which could potentially improve on the more basic approaches considered here (*Lindley, 1998; Johnson and Rossell, 2010; Morey and Rouder, 2011; Kruschke, 2018; Stahel, 2021; Micheloud and Held, 2023; Izbicki et al., 2023*).

Acknowledgments

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Conflict of interest

We declare no conflict of interest.

Software and data

The code and data to reproduce our analyses is openly available at <https://gitlab.uzh.ch/samuel.pawel/rsAbsence>. A snapshot of the repository at the time of writing is available at <https://doi.org/10.5281/zenodo.7906792>. We used the statistical programming language R version 4.3.2 (*R Core Team, 2022*) for analyses. The R packages *ggplot2* (*Wickham, 2016*), *dplyr* (*Wickham et al., 2022*), *knitr* (*Xie, 2022*), and *reporttools* (*Rufibach, 2009*) were used for plotting, data preparation, dynamic reporting, and formatting, respectively. The data from the RPCB were obtained by downloading the files from <https://github.com/mayamathur/rpcb> (commit a1e0c63) and extracting the relevant variables as indicated in the R script `preprocess-rpcb-data.R` which is available in our git repository. The RPP and EPRP data were obtained from the RProjects data set available in the R package *ReplicationSuccess* (*Held, 2020*), see the package documentation (<https://CRAN.R-project.org/package=ReplicationSuccess>) for details on data extraction.

Appendix A: Sensitivity analyses

The post-hoc specification of equivalence margins Δ and prior distribution for the SMD under the alternative H_1 is debatable. Commonly used margins in clinical research are much more stringent (*Lange and Freitag, 2005*); for instance, in oncology, a margin of $\Delta = \log(1.3)$ is commonly used for log odds/hazard ratios, whereas in bioequivalence studies a margin of $\Delta = \log(1.25)$ is the convention (*Senn, 2021*, Chapter 22). These margins would translate into margins of $\Delta = 0.14$ and $\Delta = 0.12$ on the SMD scale, respectively, using the $\text{SMD} = (\sqrt{3}/\pi) \log \text{OR}$ conversion (*Cooper et al., 2019*, p.

233). Similarly, for the Bayes factor we specified a normal unit-information prior under the alternative while other normal priors with smaller/larger standard deviations could have been considered. Here, we therefore investigate the sensitivity of our conclusions with respect to these parameters.

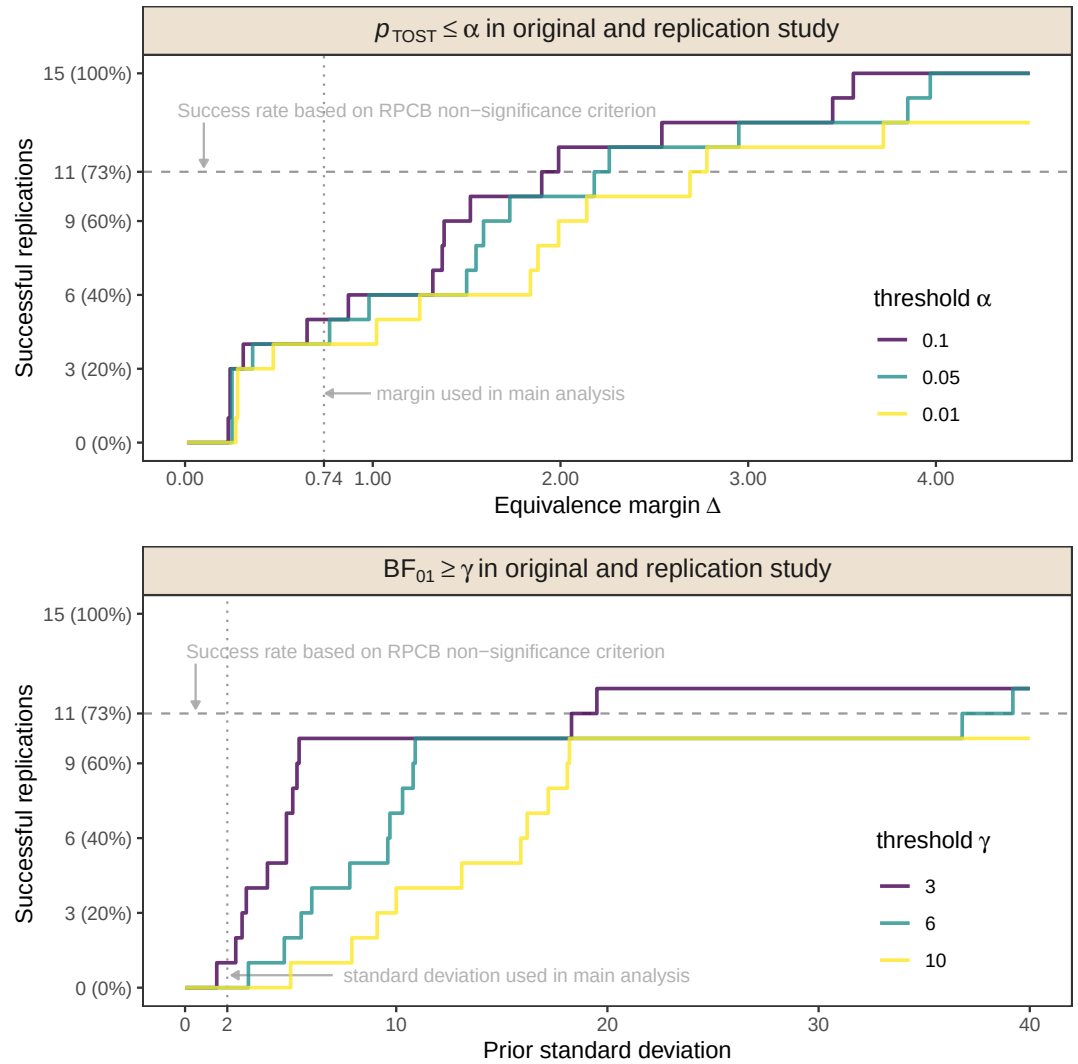


Figure 4. Number of successful replications of original null results in the RPCB as a function of the margin Δ of the equivalence test ($p_{TOST} \leq \alpha$ in both studies for $\alpha = 0.1, 0.05, 0.01$) or the standard deviation of the zero-mean normal prior distribution for the SMD effect size under the alternative H_1 of the Bayes factor test ($BF_{01} \geq \gamma$ in both studies for $\gamma = 3, 6, 10$).

The top plot of Figure 4 shows the number of successful replications as a function of the margin Δ and for different TOST p -value thresholds. Such an “equivalence curve” approach was first proposed by [Hauck and Anderson \(1986\)](#). We see that for realistic margins between 0 and 1, the proportion of replication successes remains below 50% for the conventional $\alpha = 0.05$ level. To achieve a success rate of $11/15 = 73\%$, as was achieved with the non-significance criterion from the RPCB, unrealistic margins of $\Delta > 2$ are required. Changing the success criterion to a more lenient level ($\alpha = 0.1$) or a more stringent level ($\alpha = 0.01$) hardly changes the conclusion.

The bottom plot of Figure 4 shows a sensitivity analysis regarding the choice of the prior standard deviation and the Bayes factor threshold. It is uncommon to specify prior standard deviations larger than the unit-information standard deviation of 2, as this corresponds to the assumption of

very large effect sizes under the alternatives. However, to achieve replication success for a larger proportion of replications than the observed $1/15 = 7\%$, unreasonably large prior standard deviations have to be specified. For instance, a standard deviation of roughly 5 is required to achieve replication success in 50% of the replications at a lenient Bayes factor threshold of $\gamma = 3$. The standard deviation needs to be almost 20 so that the same success rate $11/15 = 73\%$ as with the non-significance criterion is achieved. The necessary standard deviations are even higher for stricter Bayes factor thresholds, such as $\gamma = 6$ or $\gamma = 10$.

Appendix B: Null results from the RPP and EPRP

Here we perform equivalence test and Bayes factor analyses for the three original null results from the Reproducibility Project: Psychology (*Eastwick and Finkel, 2008; Ranganath and Nosek, 2008; Reynolds and Besner, 2008*) and the original null result from the Reproducibility Project: Experimental Philosophy (*McCann, 2005*). To enable comparison of effect sizes across different studies, both the RPP and the EPRP provided effect estimates as Fisher z-transformed correlations which we use in the following.

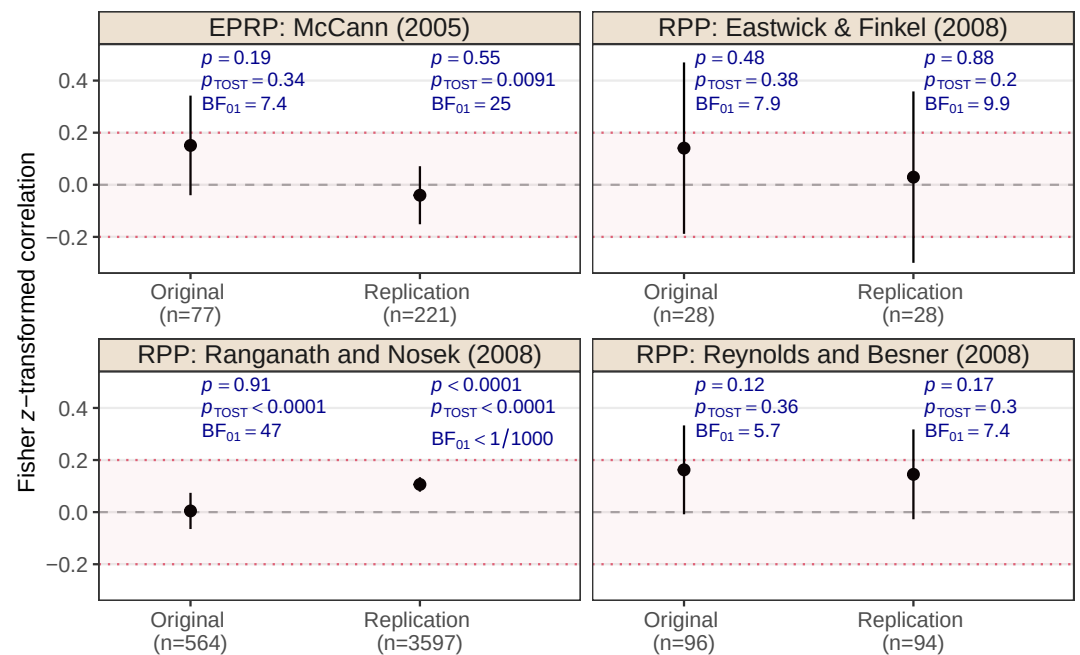


Figure 5. Effect estimates on Fisher z-transformed correlation scale with 90% confidence interval for the “null results” and their replication studies from the Reproducibility Project: Psychology (RPP, *Open Science Collaboration, 2015*) and the Experimental Philosophy Replicability Project (EPRP, *Cova et al., 2018*). The dashed gray line represents the value of no effect ($z = 0$), while the dotted red lines represent the equivalence range with a margin of $\Delta = 0.74$. The p -value p_{TOST} is the maximum of the two one-sided p -values for the null hypotheses of the effect being greater/less than $+\Delta$ and $-\Delta$, respectively. The Bayes factor BF_{01} quantifies the evidence for the null hypothesis $H_0 : z = 0$ against the alternative $H_1 : z \neq 0$ with normal prior centered around zero and standard deviation of 2 assigned to the effect size under H_1 .

Figure 5 shows effect estimates with 90% confidence intervals, two-sided p -values for the null hypothesis that the effect size is zero, TOST p -values for a margin of $\Delta = 0.2$, and Bayes factors using a normal prior centered around zero with a standard deviation of 2. We see that all replications except the replication of *Ranganath and Nosek (2008)* would be considered successful with the non-significance criterion, as the original and replication p -values are greater than 0.05. In all three cases, the Bayes factors also indicate substantial ($BF_{01} > 3$) to strong evidence ($BF_{01} > 10$) for the